

Application of Program Design to Training Seasons

The program design variables and the various types of aerobic endurance training are applied to athletes' sport seasons to create a yearly or season-long training program. Typically, the training year is divided into phases that include the off-season (sometimes called **base training**), preseason, in-season (sport competition), and postseason (active rest). [Table 21.5](#) summarizes the main objectives and the typical program design assignments for each training season.

TABLE 21.5 Sport Season Objectives and Program Design Assignments

Sport season	Objective	Frequency per week	Duration or volume	Intensity
BEGINNER ATHLETE				
Off-season (base training)	Develop sound conditioning base	3-4	Moderate to long	Low to moderate
Preseason	Improve factors important to aerobic endurance performance	3-5	Moderate to long	Low to high
In-season (competition)	Maintain factors important to aerobic endurance performance ^a	3-5 (training and racing)	Short (training)	High (training)
			Moderate	Low or high
			Race	Moderate to

^aA combination of training formats (i.e., extensive endurance, pace/tempo, and high-intensity intervals of varying distances or duration) are required to maintain aerobic endurance, LT, and exercise economy between races.

^bA combination of training formats (i.e., extensive endurance, pace/tempo, and high-intensity intervals of varying distances or duration) are required to maintain aerobic endurance, LT, and exercise economy between races.

Data from Lamb (1995); Penteado et al. (2014); Svedenhag (2008); Wells and Pate (1995).

Sport season	Objective	Frequency per week	Duration or volume	Intensity
			distance	high (racing)
Postseason (active rest)	Recovery from competitive season	2-3	Short	Low
ADVANCED ATHLETE				
Off-season (base training)	Develop sound conditioning base	5-6	Long	Low to moderate
Preseason	Improve factors important to aerobic endurance performance	6-7	Moderate to long	Moderate to high
In-season (competition)	Maintain factors important to aerobic endurance performance and increase goal race pace endurance ^b	5-6 (training and racing)	Short (training)	Low (training)
			Moderate	Low or high
			Race distance	High (racing)
Postseason (active rest)	Recovery from competitive season	3-5	Short	Low

^aA combination of training formats (i.e., extensive endurance, pace/tempo, and high-intensity intervals of varying distances or duration) are required to maintain aerobic endurance, LT, and exercise economy between races.

^bA combination of training formats (i.e., extensive endurance, pace/tempo, and high-intensity intervals of varying distances or duration) are required to maintain aerobic endurance, LT, and exercise economy between races.

Data from Lamb (1995); Penteado et al. (2014); Svedenhag (2008); Wells and Pate (1995).

Off-Season (Base Training)

The priority in **off-season** training is to develop a large base of cardiorespiratory fitness. Initially, the training program should progress to be composed of long-duration and low-intensity workouts. In the mid- to late off-season, intensity and, to a lesser extent, duration are increased; however, it is typically recommended that intermediate and advanced athletes should not increase their training duration or volume more than 5% to 10% per week.

However, for beginner athletes who start with a very low training volume, it may be safe and logical to increase by more than 10%. For example, an athlete who is new to running may run 1 mile (1.6 km) three times in their first week. By following the 10% recommendation, the following week would realize an increase of only a total of 0.3 miles (0.5 km), whereas the athlete should logically be able to absorb more than this—in this case, an additional 0.5 miles (0.8 km) on two separate days (i.e., a total of 1 mile [1.6 km] or 33% increase). Similarly, cyclists, by the nature of the volume of training they engage in as well as the low-impact nature of their sport, may be able to safely increase their volume by more than 10%. For example, a beginner cyclist may complete an average of 25 miles (40 km) in a week. The recommendation would mean the following week they would increase their volume by a total of 1.25 to 2.5 miles (2-4 km). If this athlete is training at an average speed of 10 miles per hour (16 km/h), this volume would equate to an additional 7.5 to 15 minutes, which in both absolute and relative terms is small. However, it is also important to consider that training is progressed across many weeks in a training program, and even small changes on a week-to-week basis will eventually add up to substantial increases in volume or intensity. Therefore, while an initial increase in volume of >10% may be advisable in some, continued increases across a training program of more than 5% to 10% per week may create a substantial training stress that exceeds an athlete's maximal tolerable dosage, particularly as the stress from training accumulates across the training program. Aside from background and training status, individual differences in response to training volume stimuli should also be taken into account. To avoid injury and optimize training adaptations, some athletes will require more recovery time, including a slower increase in training volume, than their peers. Monitoring subjective levels of fatigue and soreness, and objective measures such as resting heart rate and heart rate variability, can help guide appropriate increases in training volume. It should also be noted that increasing training duration too

much too quickly can actually lead to decreases in aerobic endurance performance (75).

Sample Training Program for a Beginner Marathon Runner During the Off-Season (Base Training) Phase

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Rest day	45 min Fartlek run	30 min aerobic run	Rest day	45 min aerobic run	Rest day	60 min aerobic run

Comments

- *Frequency.* To help combat overtraining or overuse, the aerobic training days should be spread out evenly during the week to allow recovery between sessions.
- *Duration.* Since the athlete's race distance is a marathon, the duration or running distance of the extensive endurance training sessions should progressively approach 80% to 85% that of the marathon, at least for one of the two aerobic (extensive endurance) sessions.
- *Intensity.* To complete the extensive endurance sessions, the athlete should run at a low intensity or a slow training pace; high respiratory stress should be avoided. The aerobic runs should be at zone 1 on the three-zone intensity scale ([figure 21.1](#) on [page 701](#)) or zone 2 on the five-zone intensity scale ([table 21.1](#) on [page 697](#)).

Sample Swim Training Program for an Intermediate 140.6 Triathlete During the Off-Season (Base Phase)

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Rest day	45 min aerobic swim	2 min intervals @ 80% $\dot{V}O_2\text{max}$ /2 min passive recovery $\times$ 8 reps	45 min aerobic swim	Rest day (no swim workout)	1 mile (1.6 km) Fartlek swim	45 min aerobic swim

This program has a swim focus; the race swim distance is 2.4 miles (3.9 km).

Comments

- *Frequency.* Because the interval and Fartlek workouts are stressful, these training sessions should be spread out during the week to allow recovery between sessions.
- *Duration.* The intervals are work bouts of 2 minutes with ≤ 2 -minute passive recovery between repetitions.
- *Repetitions.* The intervals are 6 to 10 reps $\times$ 2 minutes.
- *Intensity.* The athlete should swim at an intensity (pace) at or above 80% $\dot{V}O_2\text{max}$ when completing the work portions of the interval session. The aerobic swims should be at zone 1 on the three-zone intensity scale ([figure 21.1](#) on [page 701](#)) or zone 2 on the five-zone intensity scale ([table 21.1](#) on [page 697](#)).

Preseason

During the **preseason**, the athlete should focus on progressively increasing training intensity, maintaining or reducing duration, and incorporating all types of aerobic training stimuli into the program. The strengths and weaknesses of the individual athlete should determine the amount and frequency of each type of training.

Sample Training Program for a Beginner 50 Kilometer (31 Mile) Cyclist During the Preseason Phase

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Rest day	60 min aerobic	30 min pace/tempo	Rest day	45 min Fartlek ride	Rest day	75 min aerobic ride

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	ride	ride				

Comments

- *Frequency.* Because the pace/tempo and Fartlek rides are stressful, these two training days should be spread out during the week to allow recovery between sessions.
- *Duration.* For steady pace/tempo training, exercise duration is shorter than race distance or duration to allow for a higher training intensity.
- *Intensity.* The athlete should cycle at a high intensity or a fast training pace for both pace/tempo and Fartlek sessions; high respiratory stress is required to simulate race pace. The aerobic rides should be at zone 1 on the three-zone intensity scale ([figure 21.1](#) on [page 701](#)) or zone 2 on the five-zone intensity scale ([table 21.1](#) on [page 697](#)).

Sample Training Program for an Intermediate 10 Kilometer (6.2 Mile) Runner During the Late Preseason Phase

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Rest day	10 reps of 0.5 km (0.3 mi) intervals at race pace with a 1:1 W:R ratio	10 km (6.2 mi) easy run	45 min aerobic run	HIIT 3 min @ 90% $\dot{V}O_2$ max/2 min passive recovery $\times$ 6 reps	45 min aerobic run	45 min Fartlek run on flat course

Comments

- *Frequency.* Because the interval runs are stressful, these training sessions should be spread out during the week to allow recovery between sessions.
- *Duration.* The total distance or duration of the training portion of the session (i.e., the sum of the interval work bouts) should approach the competition distances as the athlete becomes more highly trained.
- *Intensity.* The athlete should run at an intensity (pace) close to or above LT when completing the work bout portions of the interval training sessions. The aerobic runs should be at zone 1 on the three-zone intensity scale ([figure 21.1](#) on [page 701](#)) or zone 2 on the five-zone intensity scale ([table 21.1](#) on [page 697](#)).

In-Season (Competition)

The **in-season** training program needs to be designed to include competition or race days in the training schedule. Low-intensity and short-duration training days should precede scheduled competitions so that the athlete is fully recovered and rested. The types of training employed during the in-season are based on the continued goal of improving weaknesses and maintaining strengths of the athlete.

Sample Training Program for a 5 Kilometer (3.1 Mile) Collegiate Cross-Country Runner During the In-Season Phase

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Rest or easy run	60 min aerobic run	45-min Fartlek run of hard/easy work on hills and flats	20 min pace/tempo run	30 min aerobic run with race pace pickups	20 min aerobic run with race pace pickups	Competition

Comments

- *Frequency.* Because Fartlek and tempo runs are stressful, no more than two in total should occur during the week.
- *Duration.* The total distance or duration of the training portion of the session (i.e., the sum of the interval work bouts) should approach the competition distance as the athlete becomes more highly trained.
- *Intensity.* The athlete should run at an intensity (pace) close to $\dot{V}O_2\text{max}$ when completing the work bout portions of the Fartlek training sessions and LT for the tempo run. The aerobic runs should be at zone 1 on the three-zone intensity scale ([figure 21.1](#) on [page 701](#)) or zone 2 on the five-zone intensity scale ([table 21.1](#) on [page 697](#)).

Postseason (Active Rest)

During the **postseason**, the main focus should be on recovering from the previous competitive season. Low training duration and intensity are typical for this active rest phase, but enough overall exercise or activity should be performed to maintain a sufficient level of cardiorespiratory fitness, muscular strength, and lean body mass. During the postseason, the aerobic endurance athlete should focus on rehabilitating injuries incurred during the competitive season and improving the strength of weak or underconditioned muscle groups.

A sound year-round aerobic endurance training program should be divided into sport seasons with specific goals and objectives designed to improve performance gradually and progressively.

Special Issues Related to Aerobic Endurance Training

In addition to the program design variables, it is important to consider other related issues when developing an aerobic endurance training program. These include cross-training, detraining, tapering, and supplemental resistance training. The strength and conditioning professional should contemplate these issues when adapting the types of aerobic endurance training programs to an individual athlete or developing an aerobic endurance program based on the sport season.

Cross-Training

Cross-training is a mode of training that can be used to maintain general conditioning in athletes during periods of reduced training due to injury or during recovery from a training cycle (96). Cross-training may reduce the likelihood of overuse injuries because it distributes the physical stress of training to muscle groups different from those used during training (122). Multiple-event athletes also use cross-training to maximize performance in swimming, cycling, and running. The benefits derived from cross-training include adaptations of the respiratory, cardiovascular, and musculoskeletal systems (88, 122). It seems reasonable to expect that cross-training would maintain some level of conditioning in single-event athletes who perform another mode of training (e.g., runners who perform cycling or swimming). To be effective in maintaining $\dot{V}O_2\text{max}$, cross-training must be equal in intensity and duration to the athlete's primary mode of exercise; however, cross-training will not improve single-event performance to the same magnitude as mode-specific training only (96).

Detraining

Detraining occurs when the athlete reduces the training duration or intensity or stops training altogether due to a break in the training

program, injury, or illness. In the absence of an appropriate training stimulus, the athlete experiences a loss of the physiological adaptations brought about by training. It has been demonstrated that most of the physiological adaptations attained with training regress rapidly toward pretraining levels when the training stimulus is removed (5, 102). To avoid some of the effects of detraining, the use of other training modes may be beneficial; however, cross-training may only attenuate some of the loss of physiological adaptation normally seen during complete cessation of training. Aerobic endurance athletes can minimize the effects of detraining by continuing to use their primary mode of exercise at reduced frequency and intensity, if possible (136).

Tapering

Tapering is an important component of the training program as aerobic endurance athletes prepare for major competition. Tapering involves the systematic reduction of training duration and intensity, combined with an increased emphasis on technique work and nutritional intervention. The objective of tapering the training regimen is to attain peak performance at the time of competition. While the duration of the taper depends on numerous factors, a typical tapering period may last between 7 and 28 days (91). Although most of the available research on tapering has been conducted on swimmers (95, 126, 131), the use of tapering is not limited to these aerobic endurance athletes. Research among runners and cyclists has shown that these aerobic endurance athletes also benefit from a well-planned tapering regimen (18, 91). Tapering before competition helps facilitate recovery and rehydration and promotes increases in muscle and liver glycogen stores (91).

Several types of tapering models may be employed by athletes in order to restore impaired physiological capacities resulting from the rigors of training. The most common tapering models are linear, step, and progressive tapers (131). The linear taper is characterized by a gradual decrease in the overall daily training volume throughout

the duration of the taper. In contrast, a step taper is typified by an abrupt and considerable reduction (normally $\geq 50\%$) in training volume that is maintained throughout the duration of the taper without fluctuation. The progressive taper uses a combination of the linear and step tapering models. This model is associated with a rapid 10% to 15% immediate reduction in training volume, with smaller, more gradual reductions in volume at each tier. Training volume is systematically reduced while intensity and frequency are maintained.

Concurrent Aerobic Endurance Training and Resistance Training

Resistance training is an important but often overlooked factor in improving performance in aerobic endurance athletes. This may be due to misconceptions that resistance training may increase an athlete's nonfunctional muscle mass and thereby decrease performance, or the potential interference phenomenon that may occur when resistance and aerobic endurance training sessions are completed within a training program (i.e., **concurrent training**) (35). However, apart from the well-known benefits that resistance training can provide athletes, such as reducing muscle imbalances, assistance with preventing overuse injuries, and recovery from injuries (122), numerous research investigations (25, 32, 45, 82, 84, 109, 114, 119, 127, 135), meta-analyses, and systematic review articles (1, 4, 10, 11, 17, 31, 37, 38, 65, 80, 89, 116, 143) have demonstrated that it can also improve performance (e.g., time trial, middle- and long-distance) across a number of aerobic endurance disciplines including running, cycling, cross-country skiing, and swimming, and performance-related variables such as strength, power, running economy, and maximal sprint speed. For example, a systematic review found that resistance training not only improved running performance from distances of 1,500 meters (0.9 mile) to 10,000 meters (6.2 miles) but also improved muscular strength and power factors as well as running economy (1). More specifically,

both heavy resistance training and explosive or plyometric training have been found to be effective training methods to concurrently train with aerobic endurance training to improve running economy (10, 31). Improvements in running economy and performance following heavy resistance training and plyometric training are primarily related to an increase in strength of the muscles used during running thereby reducing the relative intensity of the load of the exercising muscles (43). Plyometric training, which uses the stretch-shortening cycle, not only improves the storage and elastic energy properties of the exercising muscles but assists them to work more efficiently (86). It is important to note that maximal strength training has been found to produce greater benefits than submaximal strength training or maximal power training, with submaximal strength training producing the least gains of all other methods (10).

It is also worth highlighting that no signs of overtraining (1) or significant increases in body mass (1, 17), maximal oxygen uptake (17), or blood lactate markers (17) have been associated with the aerobic performance improvements for the majority of concurrent training studies. However, not all concurrent training studies (143) have resulted in improvements in aerobic endurance performance measures. The differences between studies is often due to the choice of resistance training exercises (e.g., high vs. low intensity, explosive or plyometric) (80), characteristics of the participants, and methods used to measure performance improvements (17). It is also worth noting that studies that reported no improvement in running economy often used a lower frequency of resistance training sessions (17) and shorter duration concurrent training programs (31, 38).

Exercise Selection

Most studies found that athletes who have benefited from concurrent training used free weight, multijoint exercises (17). This benefit may be related to the higher level of muscle coordination and

hence neuromuscular stimulus experienced with free weights compared to machine-based or single-joint exercises (17). For those studies involving running, the benefit may also be related to the greater transference from the resistance training environment to the running movement due to the biomechanical similarity between the two (17). High-velocity movements including plyometric exercises such as hops, bounds, jumps, drop jumps, and squat jumps have shown to have a positive impact on aerobic endurance performance measures (31, 38). For more information, refer to [chapters 18](#) and [19](#) for program design and technique details.

Training Intensity and Frequency

A common finding of concurrent training research for improving aerobic endurance performance is the use of heavy resistance training or plyometric training by athletes two or three times per week (10, 17) or a combination of both (4, 31, 38). However, to reduce the impact of a resistance training session on a subsequent aerobic endurance training session (i.e., residual fatigue, muscle soreness, and physiological disturbance), the volume should be reduced to limit the potential carryover of fatigue caused by the resistance training session (35). That is, keep the loads high (e.g., $\geq 90\%$ of 1RM or ≤ 4 RM) (38) but the weekly volume low (31). It is also worth noting that strength developed during a preparatory phase has been found to be maintained for up to 12 weeks with just one resistance training session per week thereafter (111), thus allowing athletes to maintain strength gains throughout subsequent training phases with minimal time commitment.

Recovery Time Between Concurrent Training Sessions

The level of recovery during a concurrent training program is influenced not only by the intensity, frequency, and duration of the individual training sessions but also the mode of aerobic endurance training (e.g., running vs. cycling) and type of resistance training performed (e.g., velocity of movement) (35). The shorter the

recovery period between training sessions, the greater the level of fatigue carried from one training session to the next (33, 110). In conjunction, there is an increase in the physiological cost of the next training session and a reduction in training performance, which may result in lower adaptations (32). Some posttraining fatigue is necessary for stimulating training adaptations, but it is simultaneously necessary to maximize recovery between training sessions, particularly if on the same day. To achieve this, manipulation of concurrent training variables such as sequence of training, training intensities, volumes, and frequency as well as the recovery period between training sessions can be adjusted (11, 34, 110). However, because athletes respond differently to training stimuli and recovery techniques, it is important to monitor athletes' recovery via measures such as heart rate, task completion times, RPE, blood lactate, power output, and the like (35).

Training Sequence

The order in which resistance and aerobic endurance training sessions are completed may affect adaptations because the acute physiological responses for resistance and aerobic endurance training are distinct to each form of training (34) and may influence the quality and quantity of training sessions completed on the same day. Findings have been mixed on the effects of concurrent training sequence on strength and $\dot{V}O_2\text{max}$ (25, 32, 45, 46). As a general guide, however, the training sequence should be varied throughout the phases of the training year and reflect the highest priority (e.g., maximizing strength or improving aerobic endurance) for each training cycle; the most important training goal should be trained first within a given workout while the athlete is fresh (125).

Mode of Aerobic Endurance Training

The results of a concurrent training meta-analysis found no significant differences across the sports studied (i.e., running, cycling, cross-country skiing, and swimming), with all sports

benefiting similarly in terms of performance from undertaking a concurrent training program (10). However, the difference in muscle action types (eccentric vs. concentric) may create a greater level of muscle damage from some modes compared to others (e.g., running vs. cycling) (143), particularly if large distances are involved. Therefore, in concurrent training programs that involve running, a greater recovery period may be needed when completed in the hours before or following a resistance training session. This concept may be especially important for athletes undertaking more than one aerobic endurance discipline, such as triathletes (35).

Athlete Training Status

The training status of athletes has been suggested to be an important variable to consider when undertaking concurrent training (11). While the majority of findings indicate that concurrent training can benefit aerobic endurance performance of athletes of different levels across different aerobic endurance disciplines (10), the responses may be different. For example, a systematic review and meta-analysis found that following a concurrent training program, trained or highly trained aerobic endurance athletes did not demonstrate any impairments to improvements in $\dot{V}O_{2\max}$; however, untrained individuals did (65). Consequently, for aerobic endurance athletes who have minimal resistance training experience or for lesser trained aerobic endurance athletes, it may be necessary to allow greater recovery times between training sessions, or apply a slower progression of training variables (e.g., intensity, frequency, or duration) or exercises compared to higher trained athletes.

[Chapter 18](#) provides guidelines for designing resistance training programs that can apply to aerobic endurance athletes; refer to scenario C for a sample program that focuses on a high school cross country runner.

Altitude

Altitude is classified into several categories, ranging from sea level (>500 m [1,640 ft]) to low (>500-2,000 m [1,640-6,562 ft]), moderate (>2,000-3,000 m [6,562-9,843 ft]), high (>3,000-5,500 m [6,562-18,045 ft]), and extreme (>5,500 m [18,045 ft]) (39). While the percentage of oxygen is the same at different altitudes (94), the atmospheric pressure drops, causing a reduction in partial pressure (PO_2), which acts as the driving force for gas exchange in the lungs (39, 94). This leads to a cascade of physiological responses to compensate for the reduction in PO_2 . Subsequently, aerobic endurance performance decrements upon acute altitude exposure may begin to occur at altitudes as low as 700 meters (2,297 ft) (145).

Acclimatization to altitude may occur between 12 and 14 days at moderate altitudes up to 2,300 meters (7,546 ft); however, it has been found that this process may take up to several months (145). According to Wyatt (145), recommendations for optimizing performance at altitude vary dramatically, from arriving immediately before competition (24-48 hr) to 12 weeks of altitude exposure.

Many elite and sub-elite athletes train at altitude to produce an ergogenic effect. In order to experience a benefit from this type of training, it has been reported that an athlete must receive a hypoxic dose of training ≥ 12 hours per day for a minimum of three weeks at moderate altitude (approximately 2,100-2,500 m [6,890-8,202 ft]) (66, 145). **Live high, train low (LHTL)** is a method commonly used by athletes seeking to benefit from altitude training. LHTL requires individuals to live at moderate altitudes, between 2,000 meters and 3,000 meters (6,562-9,843 ft), and train at or near sea level (66). This method of training allows athletes to simultaneously experience the benefits of altitude acclimatization and training at sea level (139). Thus, LHTL may potentially provide an ergogenic benefit by allowing athletes to take advantage of the metabolic and hematological adaptations experienced when living at altitude to augment neuromuscular development at lower altitudes (39, 139).

Heat Acclimation

It is well documented that aerobic endurance performances are influenced by hot environments (55, 99, 100). Consequently, undertaking training or competition in hot environments before an athlete has acclimatized results in greater thermoregulatory and cardiovascular strain, fatigue, increased perception of effort, and a negative impact on performance (52) via elevated core and skin temperatures, increased sweating, and loss of electrolytes. To avoid these negative effects, athletes are advised to acclimatize prior to competition in warm to hot conditions. Adaptations to heat acclimation protocols include an increase in plasma volume and thereby an increase in blood volume, an improvement in skin blood flow sensitivity, earlier onset of sweating at a lower core body temperature combined with an increase in sweat rate and evaporative cooling, and a reduction in gastrointestinal distress (51, 101). These changes, along with many others, help provide cardiovascular stability, lower core and skin temperatures, and a reduced exercise heart rate. A more detailed description of the adaptations to training in the heat can be found in [chapter 6](#) (see the section Heat Acclimatization).

Heat Acclimation Protocols

An athlete can adapt naturally to a hot environment by either living and training in hot ambient conditions over a long period of time (i.e., heat acclimatization) or training in an artificially hot indoor environment (i.e., heat acclimation) (108). Heat acclimation protocols are typically classified as short (<7 days), medium (8-14 days), or long term (>15 days) (50). Even though the majority (~75%) of heat adaptations occur within the first four to six days of heat exposure (e.g., heart rate and core temperature adaptations) (101), it may take seven or more days for the remainder of the adaptations to take place (e.g., enhanced core body temperature, cardiovascular, sudomotor, and sweat rate adaptations) (28, 81, 101, 108). It is recommended that athletes participating in long-duration

or multiday events as well as those of a lesser training status undertake longer protocols to ensure appropriate levels of heat adaptation (107).

The frequency of heat exposure is important because adaptations occur more rapidly with days of consecutive exposure; however, if over a 10- to 14-day protocol, one to three days of heat exposure are missed, it is unlikely that these missed days will significantly affect heat acclimation adaptations (107). It is important to note that twice-daily heat exposure is no more effective than once-daily for physiological, perceptual, or performance adaptations (141). For a heat acclimation protocol to be effective, an increase in core and skin temperatures and sweat rate is required (101, 108). To achieve this, it is generally recommended that athletes aim for 60 minutes of heat exposure (108) and a maximum of 90 minutes, although 120 minutes has been shown to be effective (115, 144). It should be noted that it may take time to build up to these exposure times over the heat acclimation protocol period as determined by the intensity of the training sessions as well as the athlete's heat tolerance. Measures of training intensity such as heart rate and performance times may be misleading in the short term until the athlete heat acclimatizes (54). A more appropriate alternative may be a scalar method such as perceived exertion (54). However, markers that an athlete is positively adapting are a lower heart rate and core temperature for a given workload, increased sweat rate, and enhanced aerobic exercise capacity (115).

Athletes with higher levels of aerobic fitness may acquire heat adaptations more rapidly and retain them longer than those athletes with a lower level of aerobic fitness (101). More aerobically fit athletes can also tolerate higher core body temperatures and therefore can perform better in the heat (101, 121). Due to the variability in responses, it is important that all athletes are monitored throughout an acclimation protocol, especially those who are just starting. Similarly, due to the added risk that hot environments can pose for athletes during training, several safety precautions should

be implemented when working with heat acclimation protocols. For example, always begin well hydrated, maintain hydration, ensure cooling equipment is available (e.g., cold-water immersion bath or ice vests), and signs and symptoms of exertional heat illness should be well known to athletes and staff and be posted. For more information, readers are directed to Pryor and colleagues (107).

To determine individual athlete adaptation and decay rates, heat acclimatization protocols should be practiced months before competition in the heat (9), and athletes should also undertake a practice heat event in similar conditions to those they expect to perform in on competition day (9). It is recommended that the environment have a wet-bulb globe temperature $>30^{\circ}\text{C}$ ($>86^{\circ}\text{F}$) to provide the necessary stress to elicit heat adaptations in core body temperature as well as induce a higher sweat response (28).

A reduction in training duration or intensity followed by a progressive increase over the protocol period may be required compared to the usual training load, on the initial introduction of a heat acclimation protocol (107). In addition, it may be necessary to use a combination of exercise and passive heat exposure (e.g., sauna or hot-water immersion) sessions as the athlete commences and progresses through the heat acclimation process to enable integration with the athlete's training schedule (54). For both hot sauna (118) and hot-water immersion (147, 148) postexercise sessions have been found to provide heat acclimation and performance benefits. However, to be successful, it is recommended that the heat stimulus be applied for a duration of 30 minutes per session, ideally on consecutive days, when possible to maximize heat acclimation adaptations (61). This process enables athletes to maintain or elevate their core body temperature and increase their sweat rate. An additional benefit of passive heat exposure is that it can reduce the volume of heat exercise training (107), which may be particularly useful when introducing unacclimatized athletes to heat acclimation protocols.

Conclusion

Training to improve aerobic endurance performance requires a well-developed and scientifically based program. The training program should be developed in conjunction with periodic performance assessment and should be structured to enhance the strengths and improve the weaknesses of the athlete. A combination of a variety of the training types described in this chapter should be used so that all physiological systems involved in successful performance are overloaded and challenged to respond with positive adaptations.

Training programs should be developed far enough in advance and with enough structure to ensure enhancement of performance but with enough flexibility to avoid overuse injuries and overtraining. Although other forms of training can be employed to avoid boredom and overtraining, activity-specific training results in the best adaptations to training and ultimately the most improvement in performance.

AEROBIC ENDURANCE TRAINING EXERCISES

The following section discusses some basic technique considerations for performance of aerobic endurance training. General guidelines for both common aerobic training machines and nonmachine activities are provided. These exercise instructions are largely adapted from Beck (8).

- 1** Treadmill [713](#)
- 2** Stationary Bike [714](#)
- 3** Rowing Machine [715](#)
- 4** Stair Stepper [716](#)
- 5** Elliptical Trainer [717](#)

- 6 Walking (Gait) [718](#)
- 7 Running (Gait) [718](#)
- 8 Freestyle Swimming [719](#)

Aerobic Endurance Training Exercises

1 TREADMILL

Starting Position

- Begin by attaching the security clip to the clothing where it will not interfere with the action of the lower or upper limbs.
- Straddle the belt by placing the feet on the right and left platform.
- Read the instructions on the console of the treadmill to gain an understanding of how to adjust the speed and incline of the specific treadmill being used.
- Turn on the machine and adjust the speed of the belt so it is at the desired warm-up speed.

Movement Phase

- While holding the handrails, allow one leg to swing freely, and using a pawing action, strike the treadmill with the mid-foot.
- Once comfortable with the speed of the belt, begin walking or running on the treadmill.
- Run or walk, keeping toward the front portion of the machine while remaining in the center of the treadmill deck.
- Release the hands from the handrails and adjust the speed and incline until the desired training level is attained.
- Avoid holding on to the console or front handrails or leaning backward while walking or running.
- Keep the head up and eyes focused forward to avoid looking down at the console for extended periods of time.

Ending Position

- Reduce the speed of the treadmill and perform a 3- to 5-minute cooldown to prevent blood pooling in the lower extremities and improve venous return to the heart.
- Step onto the platforms on either side of the belt and turn the machine off.

COMMON ERRORS

- **Continually holding the handrails—holding the handrails beyond the initial entry phase interrupts posture and stride mechanics and limits core activation; after the initial entry phase use a natural walking or running stride pattern with an upright posture and free moving arm swing, as if walking or running outside**
- **Looking continuously down at the console, which affects stride mechanics and posture and can cause neck discomfort; keep the eyes focused forward, with short glances to the console as needed to monitor display information**

2 STATIONARY BIKE

Starting Position

- Begin by adjusting the seat height so the knee of the extended leg is slightly flexed (25-30 degrees) at the bottom of the pedal stroke.
- The foot of the down leg should be flat and parallel to the floor with the balls of the feet in contact with the pedals.
- Adjust the seat so the knee is over the center of the pedal on the extended leg and the hips do not rock back and forth during pedaling.
- While maintaining a neutral spine position, lean slightly forward at the hips.

- Adjust the handlebars so that when the arms are extended at a downward angle the elbows are slightly flexed. Ideally, the lower arm and torso will form an angle of approximately 90 degrees.

Movement Phase

- Begin pedaling while keeping the balls of the feet in contact with the pedals throughout the duration of the exercise.
- Maintain a neutral spine posture and do not round the shoulders.
- With bullhorn handlebars, a variety of hand positions may be used. These include
 - pronated, palms-facing-downward grip, allowing a more upright posture;
 - neutral palms-facing grip on the sides of the handlebars, encouraging a greater forward lean; and
 - racing position, with the forearms resting on the handlebars, creating maximum forward lean.

Ending Position

- Slow down until the pedals have come to a complete stop, and then step off the bike.

COMMON ERRORS

- **If pedaling using the toes only and not the balls of the feet or moving laterally in the seat, the seat is too high. If the knees come above the height of the hips at the top of the pedaling stroke, the seat is too low.**
- **If struggling to reach the handlebars, the seat may be too far back or the handlebars too low. Correct seat and handlebar height ensures maintenance of power output throughout the pedaling stroke.**



Proper seat height adjustment: (a) leg straight with knee locked and heel on pedal; (b) knee slightly flexed with ball of foot on pedal; (c) with the pedal at 12 o'clock, the knee is about even with the hips and approximately parallel with the floor

3 ROWING MACHINE

Starting Position (Start)

- Keep the back upright, not rounded, with a slight forward lean from the hips.
- Hold the head upright while looking straight ahead.
- Extend the arms in front of the body and grab the rower handle while flexing the hips and knees until the shins are approximately vertical.

Movement Phase (Drive)

- Extend the knees first followed by the hips before using the arms to pull the handle toward the abdomen just below the rib cage. As the knees near their end range of motion, tilt the upper body backward by extending the hips before pulling with the arms to complete the movement.
- Adjust the air vent to increase (more air admitted) or decrease (less air admitted) the resistance provided.

Ending Position (Finish)

- The knees should be fully extended with the torso leaning slightly backward.
- Arms are flexed at the elbows and the handle is at the abdomen just below the rib cage.

Recovery is the movement from the finish position back to the start in the reverse order of drive phase events—that is, extension of the arms first before flexion at the hips (to move the upper body forward), then flexion of the knees back to the start position.

COMMON ERRORS

- **Using arms and legs at the same time or pulling only with arms—the legs provide the majority of the power in the drive phase; the drive phase sequence is knees, hips, then arms.**
- **The handle going up and over the knees on the way back or the chain hitting the machine in a whipping motion, which is often due to the recovery sequence being out of sync; the recovery sequence is the opposite of the drive phase, so arms, hips then knees**



Starting position (and the catch)



The drive



The finish



The recovery

4 STAIR STEPPER

Starting Position

- Hold the handrails while stepping forward onto the pedals.
- Place the whole foot in contact with each pedal.

Movement Phase

- Begin stepping, using the handrails for support.
- While maintaining an upright posture and with the legs providing the force, take deep (4- to 8-in. [10-20 cm]) steps. Keep the hips level and centered on the machine to avoid swaying from side to side or step to step.
- Do not allow the steps to contact the floor or the upper-limit stop of the machine. The heel typically comes off the pedal as

the foot is raised but should make contact again with the pedal when pressing down.

- Continue to hold the handrails lightly while looking straight ahead, keeping an upright posture with the shoulders squared and relaxed, the torso over the hips, the knees aligned with the feet, and toes pointed forward.

Ending Position

- Hold the handrails while stepping backward off the pedals.

COMMON ERRORS

- **Not keeping an upright posture, or leaning on the machine, which often is an indication that the level selected is too high**
- **Taking small steps and staying only on the toes with a high step frequency, which means the steps are too shallow**



Proper position on a stair stepper

5

ELLIPTICAL TRAINER

Starting Position

- Facing the center console of the elliptical, place one foot on each pedal.
- While standing upright and looking forward, grasp the handrails with the torso upright and evenly balanced directly over the hips, head held high, and shoulders relaxed (but not rounded).

Movement Phase

- Begin pedaling forward, moving the arms and legs in a reciprocating fashion.
- The feet should remain in full contact with the pedals throughout the duration of the exercise unless the design of the

machine causes the rear heel to lift.

- The knees should not be allowed to come forward past the toes when in the flexed position.
- Hold onto the handrails in order to maintain balance. If handrail holding is unnecessary, it is a good idea to release the hold on the handrails and pump the arms similarly to the way they move in walking or running.
- The incline of the elliptical may be increased to more closely simulate a running motion or reduced to replicate a walking motion.
- Performing this exercise with a forward motion may place greater emphasis on the quadriceps; performing it with a backward motion may increase the stress on the hamstrings and gluteals.

Ending Position

- Slow down until the machine has come to a complete stop, then step off the pedals.

COMMON ERRORS

- **Lifting the heels and applying too much pressure via the forefoot; use the whole pedal to distribute pressure when applying force to the pedals**
- **Leaning on the handrails, which disengages the core muscles and affects movement phase execution; if necessary to hold on for balance, ensure an upright posture is always maintained and the core is engaged, and avoid placing hands too low on the handrails, which can cause a forward lean**



Proper position on an elliptical trainer

6 WALKING (GAIT)

Body Position

- Hold the head upright with the eyes looking straight ahead.
- Relax the shoulders, and do not allow them to round.
- Position the upper body directly over the hips while keeping the ear, shoulder, and hip aligned.

Foot Strike

- The heel should strike the ground first, followed by a gentle rolling heel-to-ball action, allowing the weight to be spread over the foot.
- The weight should transfer from the outer side of the heel and continue to shift forward and slightly inward toward the middle of the ball of the foot at push-off.

Stride

- Without rolling the pelvis (unless racewalking), allow the hips to move freely to increase stride length.
- Lift the knees and engage the hips and gluteals in the movement.

Arm Action

- The arms should swing forward and backward in a reciprocating fashion with the lower body (i.e., when the left arm is forward, the right leg is extended, and vice versa).
- The shoulders should be relaxed, allowing the arms to swing freely.
- At faster walking speeds, the actions should be as follows:
 - The arms should be flexed at the elbows and maintained at 90 degrees with arm movement originating at the shoulders.
 - The arms and hands should swing backward and forward, not crossing the midline of the body, in order to create forward propulsion.
 - The hands should stay relaxed, with the hand coming up to the level of the chest at the nipple line on the forward swing and the hip bone at the side of the body on the backward swing.

Note: Racewalkers must increase hip rotation with each stride to create a pelvic roll. This enables them to increase stride length while keeping one foot in contact with the ground at all times per the rules of the sport.

7 RUNNING (GAIT)

Body Position

- Hold the head upright with the eyes looking straight ahead.
- Relax the shoulders, but do not allow them to round.
- Position the upper body during running with an approximately 2- to 4-degree lean from the heels to the athlete's head while keeping the ear, shoulder, and hip aligned.

Foot Strike

- The foot should strike the ground slightly in front of or directly below the athlete's center of gravity (i.e., hips), which may be a heel, mid-, or forefoot strike, followed by a gentle rolling heel-to-ball action, allowing the weight to be spread over the foot during the foot contact phase.
- The weight will likely transfer from the outer side of the heel and continue to shift.

Stride

- Without rolling the pelvis (unless racewalking), allow the hips to move freely to increase stride length.
- Lift the knees and engage the hips and gluteals in the movement.
- With each running step, the foot should land approximately under the hips to avoid braking and minimize contact time with the ground.

Arm Action

- The arms should swing forward and backward in a reciprocating motion with the lower body (i.e., when the left arm is forward, the right leg is extended, and vice versa).
- The shoulders should be relaxed, allowing the arms to swing freely.
- The arms and hands should swing backward and forward from the shoulder, not crossing the midline of the body, in order to create forward propulsion.
- Arm action of ± 20 -degree oscillation of the hands is acceptable and may improve running economy (91a).

8

FREESTYLE SWIMMING

Swimming performance is a combination of swimming-specific fitness and technique. Less-than-optimal swimming technique can significantly and negatively affect an athlete's performance due to

inefficient stroke execution, increase in drag, or a loss of power production in the water. While a number of resources are available to assist athletes and strength and conditioning professionals to address these common errors, the best method to improve swimming performance is to seek expert advice from a qualified swim coach, who can not only assess fitness and performance parameters but also advise from a technical perspective how these areas can be improved.

COMMON ERRORS

- **Swinging hand entry, in which the hand crosses over the center line on entry into the water**
- **Early hand entry, which results in a short stroke**
- **Poor catch and pull through—keeping the arm straight when pulling through the water or finishing the stroke early (i.e., not pulling through the full stroke)**
- **Ineffective breathing technique such as holding the breath until the head is out of the water or lifting the head forward to breathe**
- **Incorrect body position such as the hips dropping and therefore increasing drag due to keeping the head up to look forward**

KEY TERMS

aerobic–anaerobic interval training
age-predicted maximal heart rate (APMHR)
altitude
base training
concurrent training
cross-training
cruise intervals
detraining
duration

exercise economy
extensive endurance training
Fartlek training
frequency
functional capacity
heart rate reserve (HRR)
high-intensity interval training (HIIT)
in-season
intensity
interval training
Karvonen method
lactate threshold
live high, train low (LHTL)
long, slow distance training (LSD)
maximal aerobic capacity ($\dot{V}O_2\text{max}$)
maximal lactate steady state (MLSS)
maximum tolerated dose (MTD)
metabolic equivalent (MET)
minimum effective dose (MED)
mode
off-season
pace/tempo training
percentage of maximal heart rate (MHR) method
postseason
preseason
ratings of perceived exertion (RPE)
recovery
Tabata training
tapering
tempo intervals
threshold training
training specificity

STUDY QUESTIONS

1. Which of the following is not an adaptation following exposure to a heat acclimation protocol?
 - a. increase in cutaneous blood flow
 - b. increase in plasma volume
 - c. onset of sweating at a lower core temperature
 - d. less dilute sweat
2. All the following statements concerning heat acclimation for aerobic endurance training are correct, except for which one?
 - a. Most heat adaptations are reported to occur within the first four to six days of heat exposure.
 - b. Athletes with a lower level of aerobic fitness can acquire heat adaptations more rapidly as well as retain them longer than those athletes with a higher level of aerobic fitness.
 - c. Heat acclimatization protocols should be practiced months before competition in the heat.
 - d. Heat exposure training frequency is important because adaptations occur more rapidly with days of consecutive exposure.
3. Which of the following statements is not true concerning concurrent training?
 - a. Most studies found that athletes who have benefited from concurrent training have used machine- and single-joint-oriented exercises only.
 - b. Heavy resistance training and explosive or plyometric training by athletes two to three times per week have been found to improve aerobic endurance performance.
 - c. The level of recovery following concurrent training sessions may be influenced not only by the intensity, frequency, and duration of the individual training sessions but by the mode of aerobic endurance training and type of resistance training performed.
 - d. The training sequence should be varied throughout the phases of the training year to reflect what is most

important during that phase of training.

4. Enhanced cardiovascular and thermoregulatory function, improved mitochondrial energy production and oxidative capacity of skeletal muscle, and increased use of fat as a fuel are all physiological benefits of what type of aerobic endurance training?
 - a. intensive pace training
 - b. interval training
 - c. tempo training
 - d. extensive endurance training
5. HIIT may contribute to performance in highly trained aerobic endurance athletes via several improvements including which of the following?
 - a. improvements in peak power output
 - b. lowering of the ventilatory threshold
 - c. decreased use of fat as a fuel source
 - d. reduced hydrogen ion buffering